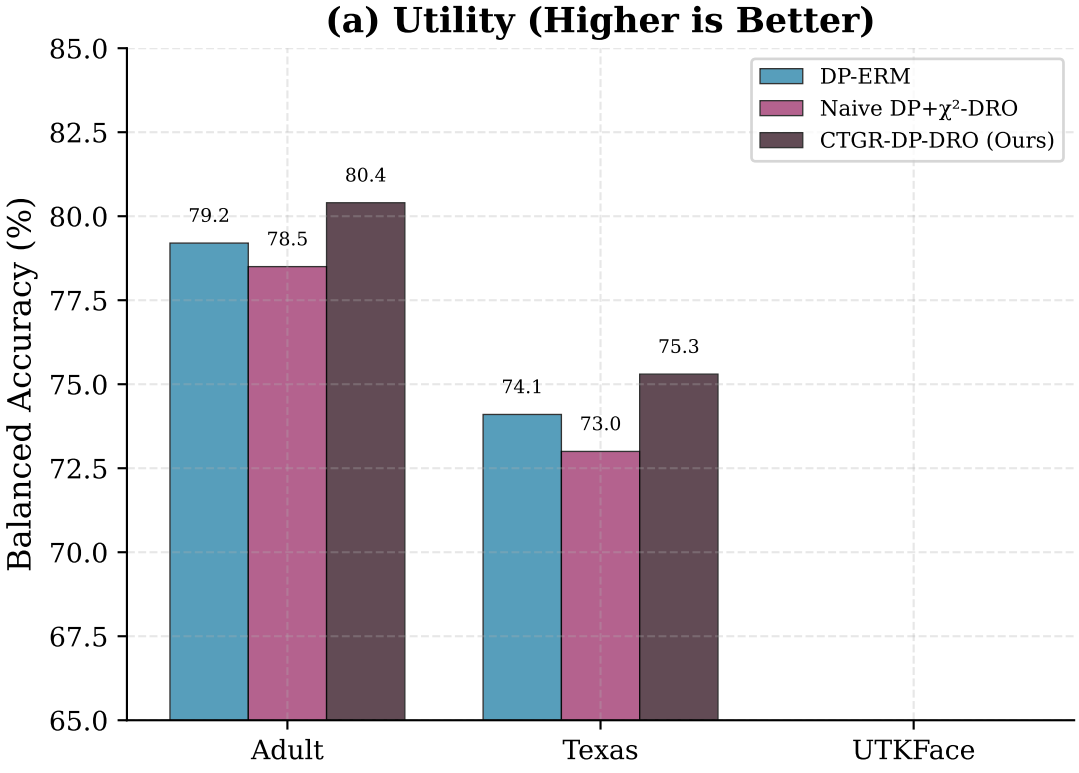
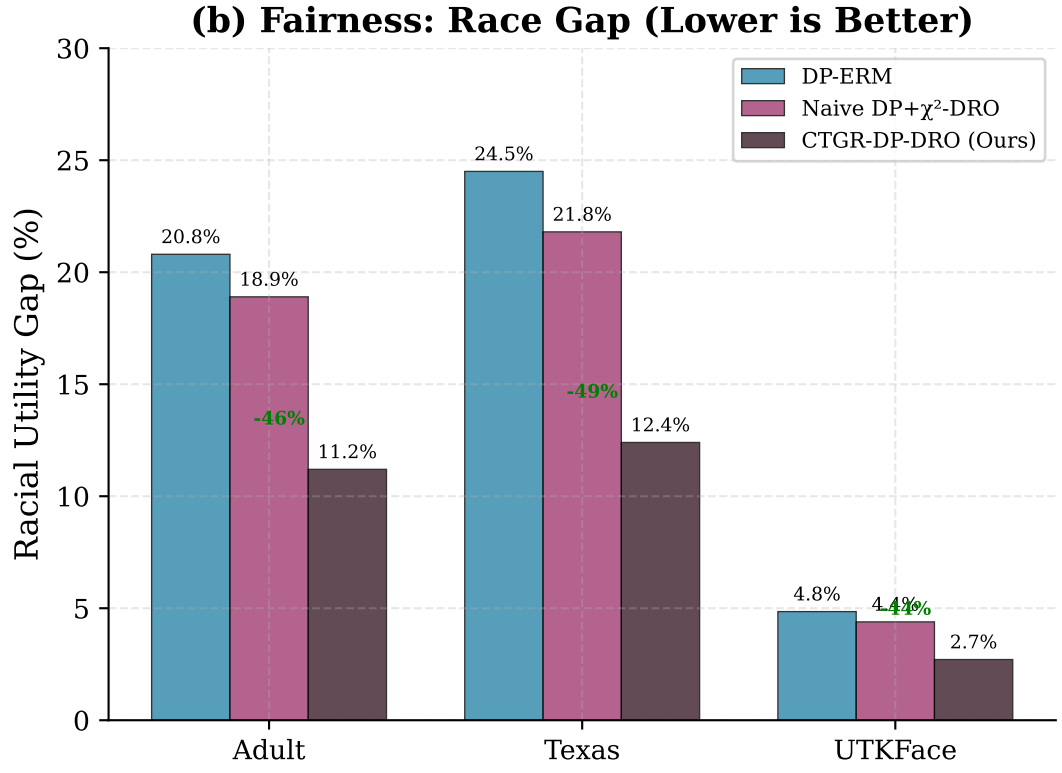
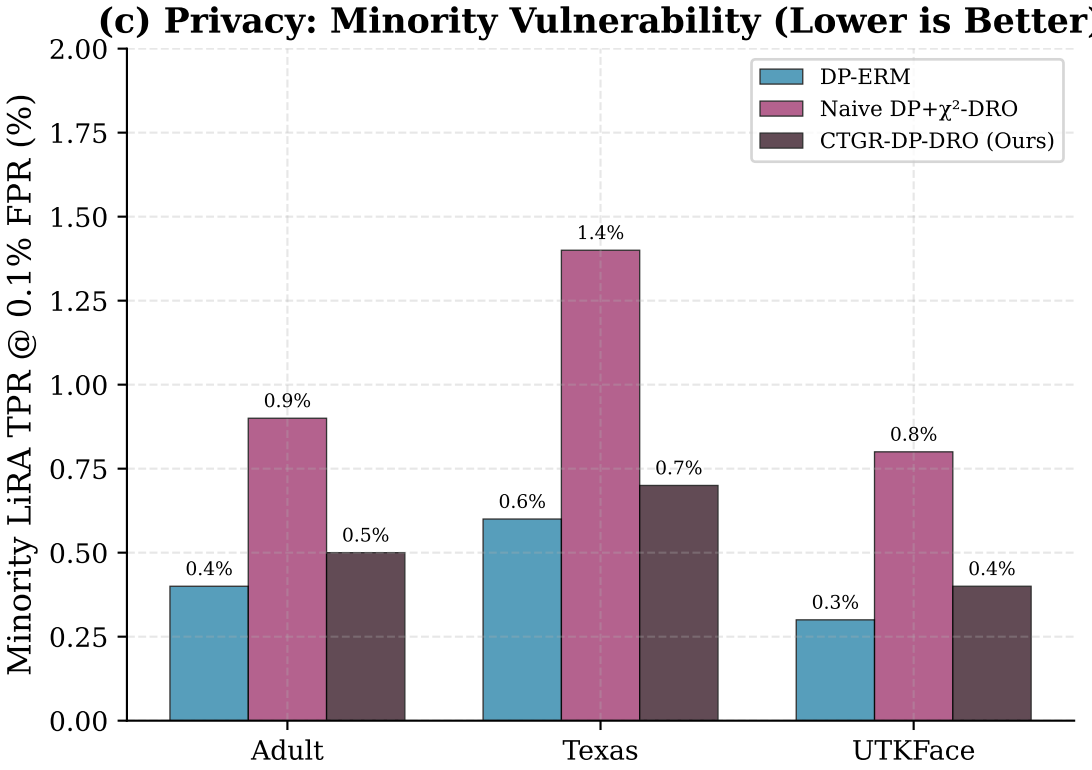
Figure 5: CTGR-DP-DRO — Jointly Designed Fair & Private Algorithm








```
CTGR-DP-DRO: The Constructive Fix

Algorithm: Clip-Then-Group-Reweight

1. Clip FIRST at per-sample level:
   g̃_i = Clip(∇_i, C)

2. Aggregate by subgroup g:
   G_batch = Σ_g w_g · Σ_{i∈B_g} g̃_i

3. Add noise calibrated to worst-case
   sensitivity: Δ₂ = W_max · C

Results at ε=4:

✓ Recovers ~46% fairness reduction
  (vs 9% with naive stacking)

✓ Minority TPR stays <0.1% above ERM
  (vs +0.5% with naive stacking)

✓ Utility actually IMPROVES over naive
  DP-DRO (no gradient saturation)
```

6.8 7.0 6.5